HealthSign

Healthcare Sign Language Communication App

Technical Documentation

Version 1.1.0

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1. Project Overview

HealthSign is a mobile application designed to bridge the communication gap between deaf or hearing-impaired patients and healthcare workers. The app supports real-time sign language gesture recognition via a custom-trained TensorFlow Lite model, speech-to-text conversion, text-to-speech output, and real-time Firebase messaging — all within a secure, role-based authentication system.

Version 1.1.0 introduces a fully custom-trained gesture recognition model. The model was trained from scratch on a combination of the ASL Alphabet dataset (Kaggle) and custom hospital-specific gesture images collected via webcam, using a TensorFlow/Keras neural network pipeline developed as part of this project.

1.1 Problem Statement

In healthcare settings, communication between hearing-impaired patients and medical staff is often difficult and inefficient. Existing solutions such as pen-and-paper or reliance on interpreters are inadequate in emergencies. HealthSign provides a real-time digital bridge using mobile AI and cloud technology to enable effective, accessible communication.

1.2 Objectives

- Enable patients to communicate using hand gestures detected via a custom-trained TFLite model
- Convert nurse speech to text and display it clearly to the patient
- Provide text-to-speech output for both parties
- Support real-time two-way messaging via Firebase Firestore
- Allow patients to send emergency call bell alerts to nearby nurses
- Train and deploy a custom gesture recognition model using ASL + hospital-specific data
- Implement a secure role-based authentication system

2. Technology Stack

Category	Technology	Purpose
Frontend	Flutter 3.x (Dart)	Cross-platform mobile UI
Backend	Firebase Firestore	Real-time NoSQL database
Authentication	Firebase Auth	Email/password + role-based access
ML Model	Custom TFLite (trained in-project)	Gesture classification from hand landmarks
Hand Detection	Google ML Kit Pose Detection	Extract body/hand landmark coordinates
Model Training	TensorFlow 2.15 / Keras	Neural network training pipeline
Landmark Extraction	MediaPipe 0.10.9	Hand landmark extraction from images
Training Dataset	ASL Alphabet (Kaggle) + Custom	87k ASL images + custom hospital gestures
Speech Input	speech_to_text	Nurse voice to text
Speech Output	flutter_tts	Read messages aloud
State Management	Provider	Reactive app state
Push Notifications	Firebase Cloud Messaging	Background nurse alerts
Cloud Functions	Firebase Functions v2 (Node.js)	FCM broadcast to nearby nurses
Location	Geolocator + Haversine	100m nurse proximity detection
Haptics	vibration	Distinct vibration patterns

3. Custom Gesture Recognition Model

A core contribution of this project is a custom-trained TensorFlow Lite gesture classification model. Unlike off-the-shelf solutions, this model was designed and trained specifically for hospital communication scenarios, using hand landmark data extracted from a combination of the publicly available ASL Alphabet dataset and a custom dataset of hospital-specific gestures recorded by the development team.

3.1 Training Pipeline Overview

End-to-End ML Pipeline

```
Step 1: Dataset Collection
  ASL Alphabet dataset (Kaggle) — 87,000 images across 29 letter classes
  Custom hospital gestures — 200 webcam photos per gesture (6 gestures)
Step 2: Landmark Extraction (extract_landmarks.py)
  MediaPipe Hands processes each image and extracts 21 hand landmarks
  Each landmark has x, y, z coordinates = 63 values per sample
  Coordinates normalised relative to wrist position for scale invariance
Step 3: Model Training (train.py)
  4-layer dense neural network trained on landmark CSV data
  Early stopping, ReduceLROnPlateau, and ModelCheckpoint callbacks
Step 4: Export (TFLite)
  Model quantised and exported to .tflite format (~50 KB)
  Deployed to Flutter app via assets/models/gesture_model.tflite
Step 5: Inference in App (gesture_service.dart)
  Camera frame -> ML Kit pose landmarks -> TFLite classifier -> gesture label
```

3.2 Dataset

Source	Details
ASL Alphabet (Kaggle)	87,000 images, 29 letter classes, used 13 letters mapped to hospital gestures
Custom Hospital Dataset	1,200 images, 6 hospital-specific gestures, recorded via webcam using collect_gestures.py

Total Samples	Approximately 7,700 after landmark extraction (some images skipped — no hand detected)
Train / Val / Test Split	70% training, 15% validation, 15% test
Input Features	63 floats per sample (21 landmarks x x, y, z coordinates)
Normalisation	All coordinates shifted relative to wrist and scaled to [-1, 1] range

3.3 Gesture Classes

ASL-based Hospital Gestures

Gesture Name	Hospital Meaning
Closed_Fist (ASL: A)	I am in pain
Open_Palm (ASL: B)	Stop / Wait
I_Feel_Cold (ASL: C)	I feel cold / shivering
Pointing_Up (ASL: D)	I need attention
I_Need_Water (ASL: I)	I need water
Thumb_Up (ASL: L)	Yes / I agree
No_Disagree (ASL: N)	No / I disagree
Need_Medicine (ASL: O)	I need medicine
Need_Restroom (ASL: R)	I need the restroom
Severe_Pain (ASL: S)	I am in severe pain
Victory (ASL: V)	I am okay
Need_Water (ASL: W)	I need water urgently
ILoveYou (ASL: Y)	Thank you

Custom Hospital Gestures (webcam collected)

Gesture Name	Hospital Meaning
Chest_Pain	I have chest pain
Headache	I have a headache
Nausea	I feel nauseous

Dizzy	I feel dizzy
Difficulty_Breathing	I have difficulty breathing
Call_Nurse	Please call a nurse

3.4 Model Architecture

Neural Network Architecture

```
Input Layer: (63,)   — 21 hand landmarks x (x, y, z)
     |
Dense(256, activation='relu')
BatchNormalization()
Dropout(0.3)
     |
Dense(128, activation='relu')
BatchNormalization()
Dropout(0.3)
     |
Dense(64, activation='relu')
BatchNormalization()
Dropout(0.2)
     |
Dense(32, activation='relu')
Dropout(0.1)
     |
Dense(19, activation='softmax')  — 19 gesture classes
     |
Output: probability distribution over 19 classes
```

3.5 Training Configuration

Parameter	Value
Optimizer	Adam (learning rate = 0.001)
Loss function	Sparse categorical cross-entropy
Batch size	32
Max epochs	100

Early stopping	patience = 15, monitors val_accuracy
LR scheduler	ReduceLROnPlateau — halves LR if val_loss stagnates for 5 epochs
Best model saving	ModelCheckpoint saves best val_accuracy weights
Model export	TFLite with DEFAULT quantisation optimisation
Final model size	~50 KB

3.6 Training Scripts

Script	Purpose
collect_gestures.py	Opens webcam to record 200 photos per custom hospital gesture with live countdown and progress bar
extract_landmarks.py	Processes all dataset images with MediaPipe, extracts 21 hand landmarks per image, normalises and saves to landmarks.csv
train.py	Trains neural network, generates timestamped training curve + confusion matrix plots, exports TFLite model
test_live.py	Opens webcam and runs the trained TFLite model in real time — shows gesture name, hospital meaning, and confidence bar

3.7 Model Performance

Expected Results After Training

```
Test Accuracy          : 90 - 96%
Model size             : ~50 KB (quantised TFLite)
Inference time         : < 5ms per frame
Input shape            : (63,)
Output shape           : (19,) softmax
Gesture classes        : 19
Training plots saved   : plots/training_curves_TIMESTAMP.png
```

3.8 Integration in Flutter App

The trained gesture_model.tflite file is placed in the Flutter app's assets folder and loaded at runtime by the GestureService class. The inference pipeline works as follows:

In-App Inference Pipeline

1. CameraController streams frames (NV21 format, low resolution)
2. Frame converted to InputImage with correct rotation metadata
3. ML Kit PoseDetector.processImage() extracts body landmark coordinates
4. Wrist, elbow, shoulder positions extracted as 63-float vector
5. TFLite interpreter runs gesture_model.tflite on the vector
6. Softmax output gives probability for each of 19 gesture classes
7. Stability filter: same gesture must appear 4 consecutive frames
8. If confidence >= 75%, gesture label + text sent to Firestore + TTS

4. System Architecture

4.1 High-Level Architecture

Architecture Layers

UI Layer (Screens & Widgets)

Patient Home → Nurse Home → Auth Screens → Gesture Reference → History

State Management (Providers)

AuthProvider → ConversationProvider → ChatRequestProvider

Services Layer

AuthService → FirestoreService → SpeechService → GestureService (TFLite) → HapticService → NotificationService

ML Layer

Custom TFLite Model (gesture_model.tflite) + ML Kit Pose Detection

Cloud Layer (Firebase)

Auth → Firestore → Cloud Messaging → Cloud Functions (Node.js v2)

4.2 Folder Structure

Project Directory

```
lib/config/          app_theme.dart, app_constants.dart
lib/models/          user_model.dart, message_model.dart,
chat_request_model.dart
lib/providers/       auth_provider.dart, conversation_provider.dart,
chat_request_provider.dart
lib/services/        auth_service.dart, firestore_service.dart,
speech_service.dart,
                    gesture_service.dart (TFLite), haptic_service.dart,
notification_service.dart
lib/screens/         auth/ → patient/ → nurse/ → shared/
assets/models/      gesture_model.tflite, gesture_labels.txt
ml_training/         collect_gestures.py, extract_landmarks.py, train.py,
test_live.py
functions/           index.js (Firebase Cloud Functions)
```

5. Firebase Database Structure

5.1 Collections

users

Field	Type / Description
uid	String — Firebase Auth UID (document ID)
name	String — Full name
email	String — Email address
role	String — 'patient' or 'nurse'
createdAt	Timestamp — Account creation time
fcmToken	String — FCM push notification token
location	GeoPoint — GPS coordinates for 100m proximity
isAvailable	Boolean — Whether nurse is available to receive calls

conversations

Field	Type / Description
patientId	String — UID of the patient
createdAt	Timestamp — Conversation start time
lastMessage	String — Preview of last message
lastUpdated	Timestamp — Time of last message

conversations/{id}/messages

Field	Type / Description
conversationId	String — Parent conversation ID
senderId	String — UID of sender
senderRole	String — 'patient' or 'nurse'
text	String — Message content
type	String — 'text', 'gesture', 'speech', or 'shortcut'
timestamp	Timestamp — Time sent

isRead	Boolean — Read receipt

chat_requests

Field	Type / Description
nurseId	String? — UID of responding nurse (null if patient-initiated)
nurseName	String — Display name of nurse
patientId	String — UID of patient
patientName	String — Display name of patient
status	String — 'pending', 'calling', 'accepted', 'declined', or 'ended'
urgency	String — 'normal' or 'emergency'
initiatedBy	String — 'patient' or 'nurse'
createdAt	Timestamp — When request was created
endedAt	Timestamp? — When conversation ended
conversationId	String? — Linked conversation after acceptance
declineReason	String? — Reason if nurse declined
patientLocation	GeoPoint? — Patient GPS for proximity targeting
nearbyNurseIds	Array — UIDs of nurses within 100m

6. Core Features

6.1 Gesture Detection (Custom TFLite Model)

The gesture detection pipeline uses a custom-trained TensorFlow Lite model deployed within the Flutter app. Camera frames are processed by ML Kit Pose Detection to extract body landmark coordinates, which are passed to the TFLite model for classification.

Feature	Implementation
Model	Custom-trained TFLite — gesture_model.tflite (~50 KB)
Hand detection	Google ML Kit Pose Detection (body landmark extraction)
Input	63 floats — 21 pose landmarks x (x, y, z) normalised to wrist
Output	Softmax over 19 gesture classes
Stability filter	4 consecutive identical detections before accepting gesture
Confidence threshold	75% minimum confidence before gesture is accepted
Supported gestures	19 total — 13 ASL-mapped + 6 custom hospital gestures
Processing	Async in camera image stream — does not block UI thread

6.2 Chat Request System

Both patients and nurses can initiate communication. Patients press a call bell to alert nearby nurses. Nurses can also search for and select a specific patient to start a conversation.

Two-Way Chat Initiation

Patient -> Nurse (Call Bell):

Patient taps call bell -> Normal or Emergency request saved to Firestore
Firebase Cloud Function detects new document -> sends FCM to nearby nurses (100m)
Nurse sees animated banner on ALL tabs -> Answer or Decline with reason

Nurse -> Patient (Ward Round):

Nurse searches patient list by name -> taps Chat button
Patient receives popup with nurse name -> Accept or Decline

6.3 Push Notifications

Scenario	Implementation
Patient sends call bell	Firebase Cloud Function (v2) triggers on Firestore create, sends FCM multicast to all available nurses within 100m
App in foreground	flutter_local_notifications shows heads-up notification banner
App in background	FCM delivers notification via Android system notification tray
App killed/closed	FCM wakes app via getInitialMessage() handler on next open
Location targeting	Haversine formula calculates distance between patient and nurse GPS coordinates
Emergency priority	Android FCM priority set to max with urgent vibration pattern

6.4 Haptic Feedback

Event	Vibration Pattern
Incoming nurse request	3 strong pulses: 300ms ON, 150ms OFF x 3
New message received	2 soft taps: 80ms ON, 80ms OFF x 2
Request accepted	Escalating: 50ms, 100ms, 200ms
Request declined	Single long buzz: 500ms

7. Screen Reference

Screen	Description
SplashScreen	Animated loading. Checks Firebase auth state and routes to correct home screen.
LoginScreen	Email + password with validation. Routes to patient or nurse home based on role.
RegisterScreen	Name, email, password, role selector. Creates Firebase Auth user + Firestore document.
PatientHomeScreen	Camera panel with TFLite gesture detection, quick shortcuts bar, message list, call bell button.
NurseHomeScreen	Salomon bottom nav (Chat / Patients). Global call banner overlays all tabs.
Nurse Chat Tab	STT microphone panel, message list with auto-scroll, input bar, end conversation button.
Nurse Patients Tab	Searchable patient list, incoming call bell cards with Answer/Decline, patient call history.
GestureDemoScreen	Reference grid of all 19 supported gestures with Lottie animations. Tap to hear via TTS.
ConversationHistoryScreen	Messages grouped by date for patient side.
NurseChatHistoryScreen	Ended conversations list for nurse side — streamed from Firestore in real time.

8. Build & Deployment

8.1 Android Requirements

Requirement	Version
Flutter SDK	3.x stable
compileSdk	36
minSdk	26 (Android 8.0)
targetSdk	36
Kotlin	2.2.20
Java	17
Core library desugaring	com.android.tools:desugar_jdk_libs:2.1.4

8.2 Build Commands

Build Commands

```
# Debug run on emulator
flutter run

# Release APK
flutter build apk --release

# Deploy Cloud Functions
firebase deploy --only functions

# Train ML model (in ml_training/)
python collect_gestures.py   # record custom gestures
python extract_landmarks.py  # extract hand landmarks
python train.py              # train neural network
python test_live.py          # test with webcam
```

9. Android Permissions

Permission	Purpose
CAMERA	Live camera feed for gesture detection
RECORD_AUDIO	Microphone input for speech-to-text
INTERNET	Firebase and FCM communication
VIBRATE	Haptic feedback for notifications
MODIFY_AUDIO_SETTINGS	TTS audio output control
ACCESS_FINE_LOCATION	GPS coordinates for 100m nurse proximity targeting
ACCESS_COARSE_LOCATION	Fallback location for proximity
POST_NOTIFICATIONS	Android 13+ push notification permission

10. Known Limitations & Future Work

10.1 Current Limitations

- Gesture detection uses body pose landmarks (wrist/elbow/shoulder) rather than finger-level landmarks, limiting the precision of gesture classification
- Push notifications require both devices to have granted notification permission and the nurse must have opened the app at least once for FCM token registration
- The 100m proximity radius requires location services to be enabled on the nurse device
- Speech-to-text requires an active internet connection
- The app currently supports one active conversation per nurse at a time

10.2 Future Enhancements

- Integrate MediaPipe Hand Landmark Detection for finger-level gesture classification (21-point hand skeleton)
- Expand the custom training dataset with more hospital workers to improve model generalisation
- Add video call support for real-time visual communication
- Support multiple simultaneous conversations for nurses
- iOS platform support
- Offline mode using Firestore local persistence
- Multi-language TTS and STT support for multilingual hospital environments

11. Dependencies

Package	Version	Purpose
firebase_core	^3.6.0	Firebase SDK initialisation
firebase_auth	^5.3.1	User authentication
cloud_firestore	^5.4.4	Real-time database
firebase_messaging	^15.1.3	Push notifications (FCM)
flutter_local_notifications	^18.0.1	Foreground notification display
provider	^6.1.2	State management
camera	^0.11.0+2	Camera access and image stream
google_mlkit_pose_detection	^0.11.0	Body landmark detection
tflite_flutter	^0.10.4	TFLite model inference
speech_to_text	^7.0.0	Microphone speech recognition
flutter_tts	^4.0.2	Text-to-speech output
geolocator	^13.0.1	GPS location for proximity
permission_handler	^11.3.1	Runtime permissions
vibration	^2.0.0	Haptic feedback
salomon_bottom_bar	^3.3.3+1	Animated bottom navigation
lottie	^3.1.2	Animated gesture icons
google_fonts	^6.2.1	Inter font family
intl	^0.19.0	Date/time formatting
uuid	^4.5.1	Unique message IDs

12. Glossary

Term	Definition
TFLite	TensorFlow Lite — lightweight ML inference format for mobile devices
TTS	Text-to-Speech — converts written text to spoken audio
STT	Speech-to-Text — converts spoken audio to written text
ML Kit	Google's on-device machine learning SDK for mobile apps
MediaPipe	Google's framework for processing perceptual data including hand and pose landmarks
ASL	American Sign Language — used as source dataset for hand shape training data
Landmark	A specific point on a body or hand detected by ML Kit (x, y, z coordinates)
TFLite Quantisation	Reduces model precision from float32 to int8 to shrink file size with minimal accuracy loss
Softmax	Neural network output function that converts raw scores to probabilities summing to 1.0
Firestore	Google Cloud Firestore — NoSQL real-time cloud database
FCM	Firebase Cloud Messaging — push notification delivery service
Haversine	Mathematical formula for calculating geographic distance from latitude/longitude coordinates
Provider	Flutter state management pattern using InheritedWidget
APK	Android Package Kit — installable file format for Android apps
Confiden ce	Probability score (0.0–1.0) output by the model for the predicted gesture class
Early Stopping	Training technique that stops when validation accuracy stops improving to prevent overfitting
Overfitting	When a model memorises training data and performs poorly on unseen data
Batch Normalisation	Technique that stabilises and speeds up neural network training
Dropout	Regularisation technique that randomly deactivates neurons during training to prevent overfitting